Dominic Irla

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Education

University of Illinois at Chicago Chicago, IL

Bachelor of Science in Computer Science *Expected May 2027*

- 3.85 GPA. Coursework includes Software Design, Algorithmic Practices, WebGPU, and UX Design. Dean's List.

Technical Skills

Languages: Python, JavaScript/TypeScript, SQL, Java, C/C++, Go, HTML/CSS
Frameworks: React, Node.js, Laravel, Tailwind, Vite, Astro
Infrastructure & Data: PostgreSQL, REST APIs, Firebase, Supabase, TikTokCloud, Consul

Work Experience

ByteDance / TikTok May 2026 – August 2026

Frontend Software Engineer Intern – Global Payment Team

- Built **H5** web features for TikTok's global **BNPL** payment product across SEA and MX/BR markets, implementing timezone conversion, GPS/location probing, monitoring & alerting, and geo-restriction logic.
- Shipped updates across other payment products (refund system, material-type auditing) and extended frontend work to ∼10 screens in the TikTok Shop Android client, on a 10-person team collaborating internationally with business and backend engineers.
- Deployed features on **TikTokCloud** infrastructure, integrating with **Consul** for service discovery.
- Built an **AI-powered** internal reporting tool consolidating 50+ release alerts into a single report, and an internal debug tool for generating mock/fake QA users.

UIC College of Engineering June 2024 – Present

Web Developer – Full-Stack

- Designed and built **COEConnect**, a unified internal portal/SDK replacing duplicated **Shibboleth (SSO)** auth, mail, and ops integrations across **∼10 department apps**, with a **PostgreSQL**-backed dashboard for live health monitoring, observability, and centralized logging.
- Lead a team of 4 rebasing legacy internal apps onto a modern **React** stack, resolving technical debt and improving maintainability.

Extracurricular Activity

UIC Research – Wearable Technology & Sensory Environment Lab March 2026 – Present

Web Developer

- Rebuilt the frontend of a serious-games research project (**React/TypeScript, Vite**) enabling patients to interact with therapeutic games via a custom tongue trackpad, rendered live via canvas.
- Integrated trackpad hardware over **Web Bluetooth (BLE/GATT)**, streaming live sensor data for 12 study patients with no backend, handling pairing, subscriptions, and parsing under strict client-side-only **HIPAA** constraints.
- Engineered **PSoC**-based embedded firmware for the trackpad's capacitive sensing, reducing packet loss by **3x** and increasing **CapSense** resolution by **4x**.

Sparkhacks July 2025 – Present

Web Development Lead

- Lead a team of 6 building the application portal for Chicago's largest hackathon (**400+ entries**) using **Astro and Firebase**, supporting ∼500 dashboard users.

Association for Computing Machinery – UIC Chapter Sept 2025 – Present

SIG Lead – Web Development

- Lead a team building **LinkIt**, a **React/Supabase** real-time app; run weekly workshops on frontend frameworks and REST APIs; mentor members on code quality and system design.

Projects

Automated Drone Vision | Python, Embedded Systems, YOLO August 2025

- Wrote **Python** for an **Nvidia Jetson**-controlled drone to autonomously identify and retrieve targets using **YOLO** object detection.

Awards / Certificates

1st Place — UX Design Contest | FBLA, IL July 2023

- Earned 1st place at the state level for applied UX design and interface design.